

IoT Temperature Monitoring & Automatic Motor Control

AGENDA

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ABSTRACT

Temperature monitoring and timely control of equipment are important for human safety, equipment protection, and operational reliability across industrial, healthcare, environmental, and automation applications. This project demonstrates an IoT-based real-time temperature monitoring and automatic motor control system using an LM35 sensor, Arduino UNO, ESP8266, MQTT, and ThingSpeak, validated through Proteus 9.1 VSM simulation.

The system connects physical sensing with cloud-based monitoring and automated decision-making. Temperature data is transmitted to ThingSpeak, where a predefined threshold triggers an automatic motor-control action when the temperature exceeds 28°C.

The project's global significance lies in demonstrating how IoT can transform conventional monitoring into connected, remotely accessible, and intelligent systems, providing a foundation for safer operations, faster responses, and scalable automation in real-world applications.

PROJECT OUTCOME

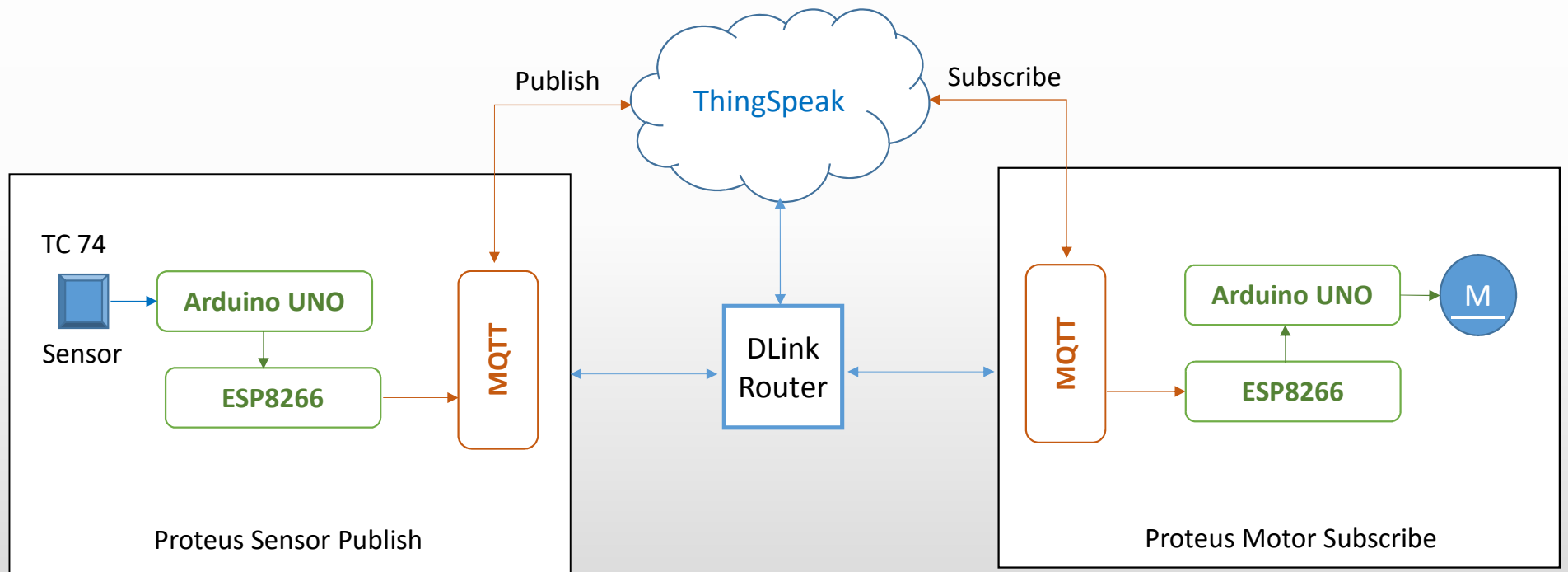
- Successfully demonstrated real-time temperature monitoring using an TC74 sensor through Arduino UNO and ESP8266.
- Established reliable MQTT-based communication between the simulated embedded system and ThingSpeak Cloud.
- Implemented automatic motor control based on a predefined temperature threshold of 28°C.
- Integrated ThingSpeak React and ThingHTTP for cloud-based decision-making and control.
- Validated the complete sensor-to-cloud-to-actuator IoT workflow using Proteus 9.1 VSM simulation.
- Applied systematic MQTT, DNS, API, and configuration troubleshooting to resolve communication and control issues.
- Demonstrated a scalable foundation for remote monitoring, intelligent automation, and real-world IoT applications.

PROJECT IMPACT

- Demonstrates how IoT technology can convert conventional monitoring into connected and automated systems.
- Enables real-time remote temperature monitoring through cloud-based data visualization.
- Reduces dependence on manual monitoring and intervention through automatic motor control.
- Supports faster response to abnormal temperature conditions, improving operational reliability.
- Provides a foundation for applications in industrial automation, healthcare, environmental monitoring, and equipment protection.
- Demonstrates a scalable IoT architecture that can be extended with additional sensors, alerts, analytics, and cloud platforms.
- Highlights the potential of MQTT and cloud technologies for building cost-effective and remotely accessible IoT solutions.

BLOCK DIAGRAM

IoT Based Temperature Monitoring & Automatic Motor Control System



BRIEF DESCRIPTION OF EACH BLOCK

Hardware:

It comprises the Arduino UNO, TC74 Sensor, Motor shield.

ESP8266:

It works as a Gateway to process the data and send it to the Cloud Platform. Here Dlink router is providing the Internet connection.

Software:

Proteus Design Suit

ThingSpeak:

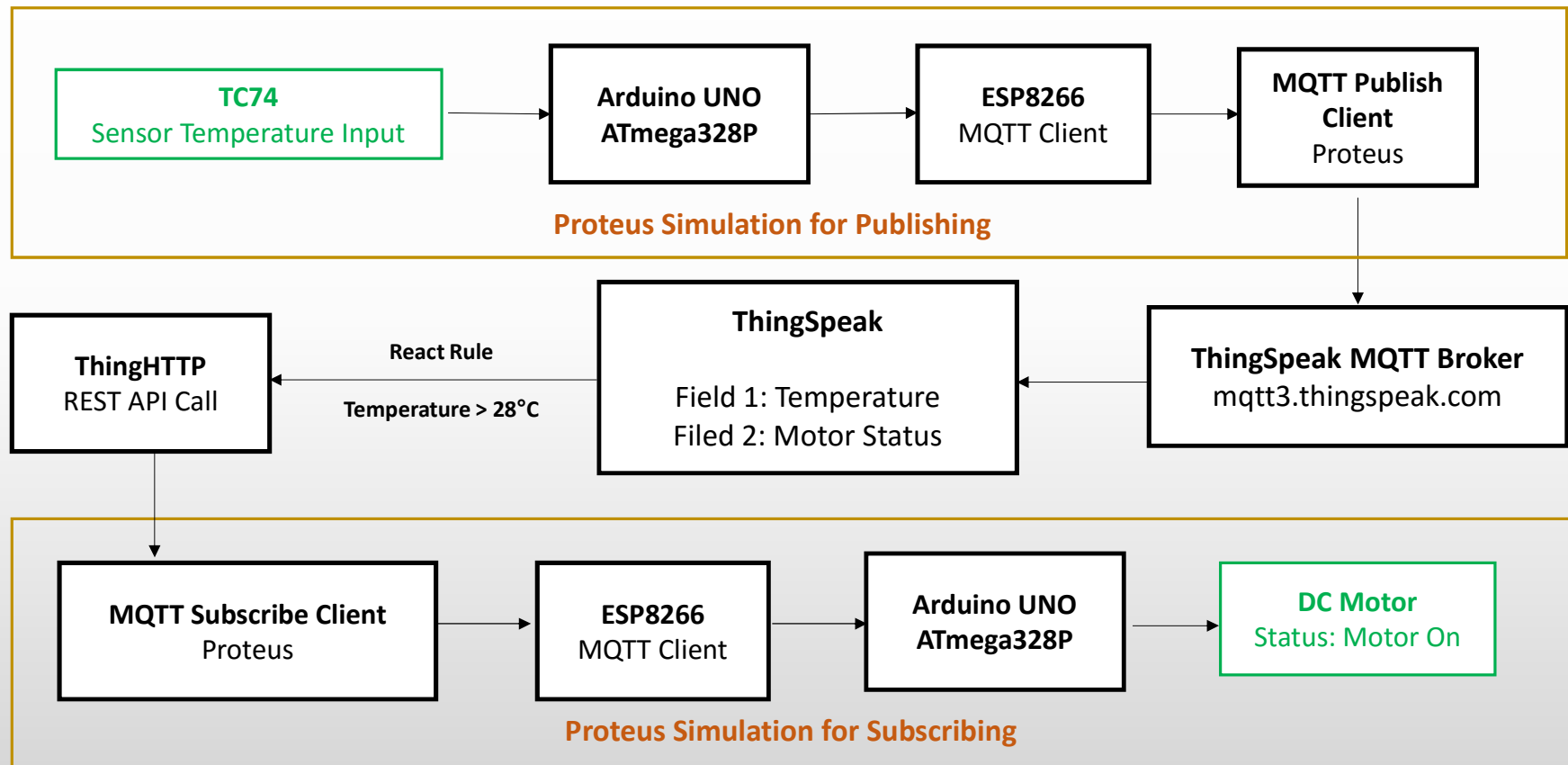
It is a open source cloud platform where we can store our sensor data for storage and threshold limit checking about the temperature reading and based on the condition activate motor control.

How this Prototype works:

TC74 sensors collect the relative parameters from the surrounding air and capture the temperature in the Arduino UNO board.

This data is then send to the cloud platform (ThingSpeak) for storage and threshold limit checking and based on the condition it sends command to activate motor.

FLOW CHART OF TEMPERATURE MONITORING & AUTOMATIC MOTOR CONTROL SYSTEM



TOOLS USED

Hardware	Simulation Component
Microcontroller	• Arduino UNO • ESP8266
Sensor	• TC74 temperature sensor • Connected to Arduino using I2C
Communication	• ESP8266 Wi-Fi module • Proteus ESP1:MQTT component • MQTT communication with ThingSpeak
Actuator	• DC Motor Shield • Motor runs in FORWARDS direction at speed 255 when the ON command is received

TOOLS USED

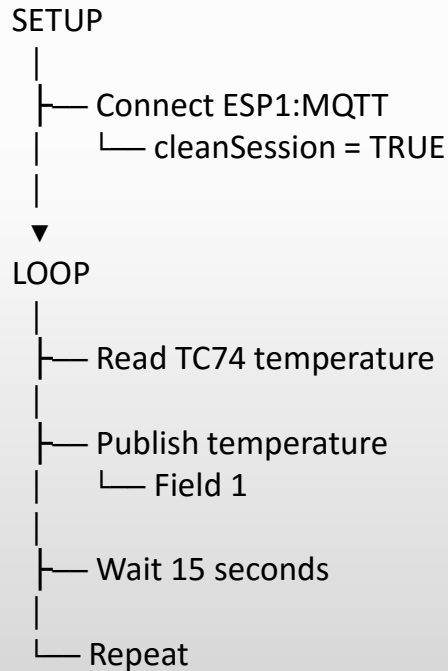
Software tool/ Platform	Purpose
Proteus 9.1 VSM	Embedded system and IoT simulation
Proteus Visual Designer	Firmware flowchart development
ThingSpeak	Cloud platform, dashboard and automation
MQTT	IoT communication protocol
Mosquitto	MQTT Broker
mosquitto_pub	MQTT publishing
mosquitto_sub	MQTT subscription
ThingSpeak React	Cloud-based rule engine
ThingHTTP	REST API based cloud action

EXPERIMENTAL SETUP

Two independent Proteus projects are used.

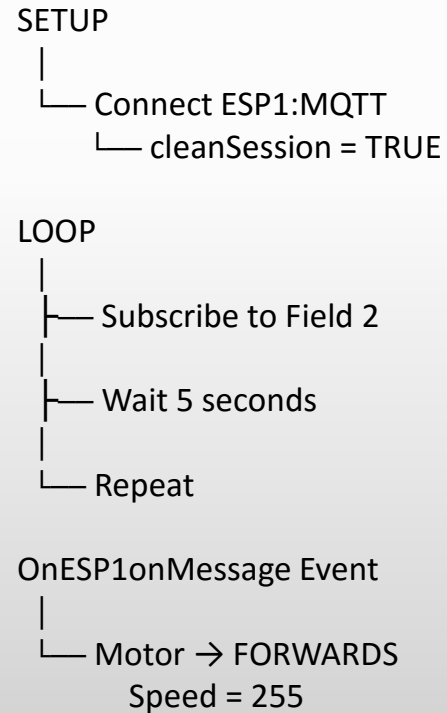
1. Temperature Publish Project

The publish project reads the TC74 temperature and sends the value to ThingSpeak Field 1.



2. Motor Subscribe Project

The subscribe project continuously listens for motor-control commands.



EXPERIMENTAL SETUP

MQTT Communication

MQTT Broker

Broker: mqtt3.thingspeak.com
Port: 1883
QoS: 0
Clean Session: 1

Temperature Publish Topic

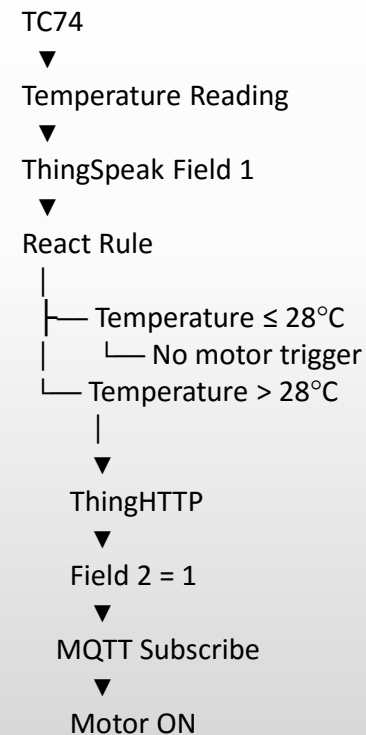
channels/CHANNELID/publish/fields/field1

Motor Subscribe Topic

channels/CHANNELID/subscribe/fields/field2

Temperature-to-Motor Control Logic

The automatic control sequence is:



EXPERIMENTAL SETUP

ThingSpeak Configuration

The ThingSpeak implementation uses:

- Field 1: Temperature
- Field 2: Motor Status
- React: Temperature threshold monitoring
- ThingHTTP: Field 2 update
- Dashboard: Gauge, line chart and motor LED

Two separate MQTT devices are configured:
Separate devices are used to avoid MQTT Client ID conflicts.

Proteus_publish

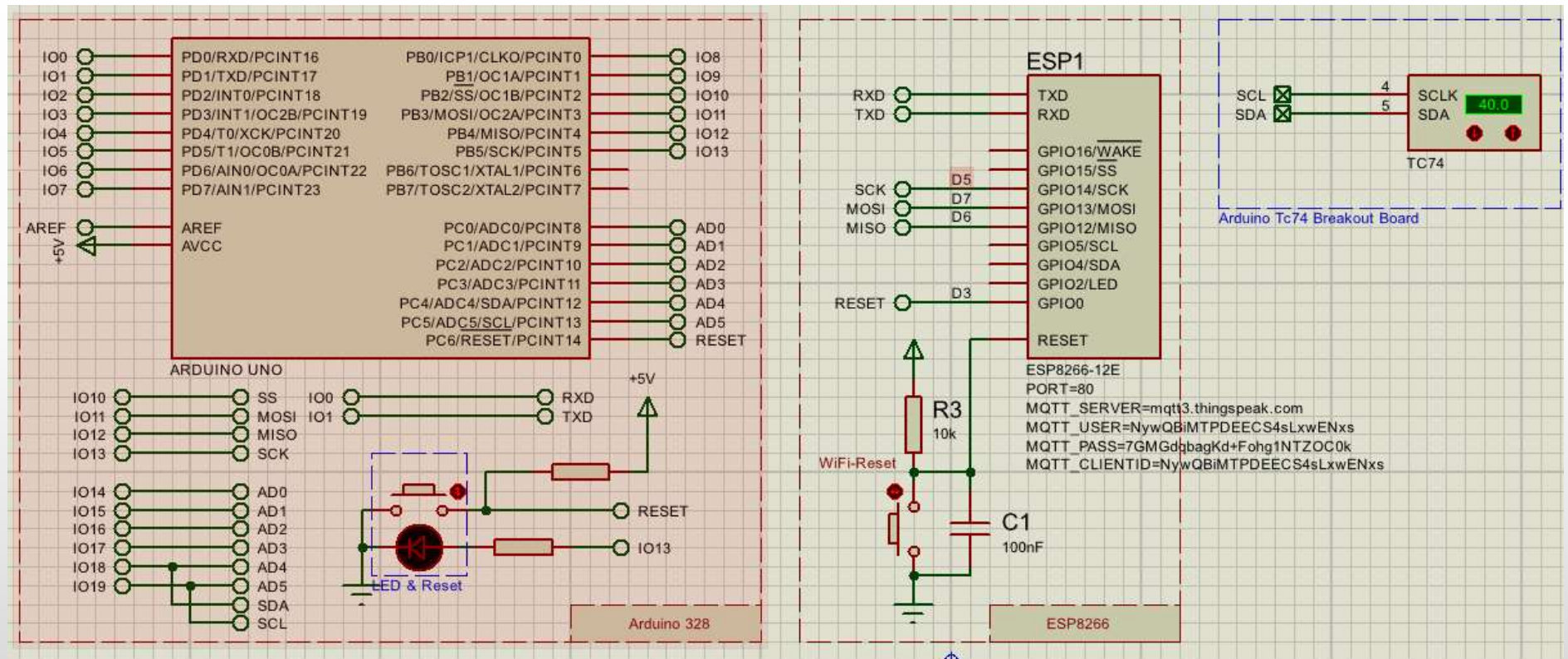
Function: Publishes temperature data

Proteus_subscribe

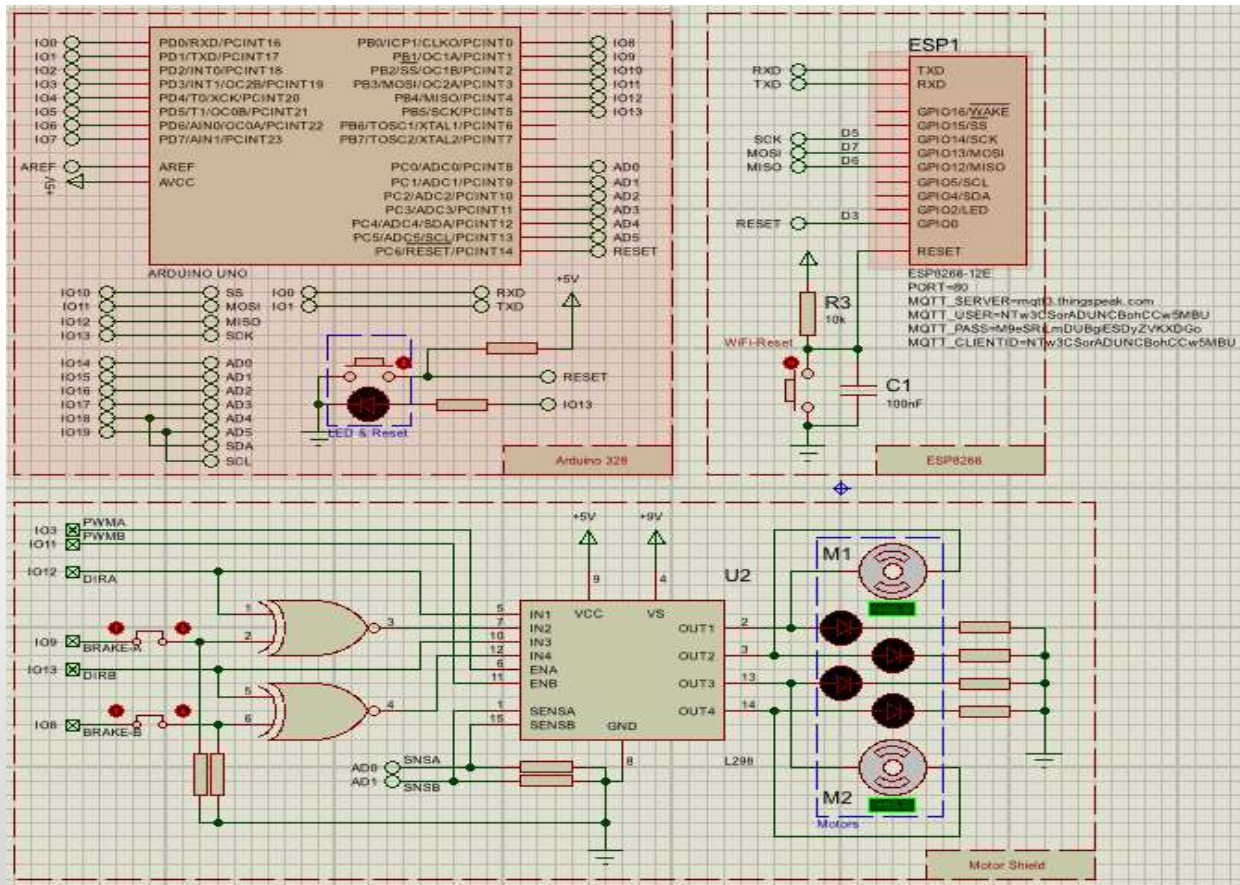
Function: Subscribes to motor-control commands

SCHEMATIC IN PROTEUS

for Publishing the Sensor Value



for Subscribing to Activate Motor



TESTING & VALIDATION

The complete system was tested using both command-line MQTT tools and Proteus simulation.

Test	Expected Result	Result
DNS resolution	Broker IP returned	Passed
MQTT connection	CONNACK received	Passed
Temperature publish	Field 1 updated	Passed
React rule	Field 2 updated when temperature > 28°C	Passed
Motor subscribe	Motor command received	Passed
End-to-end control	Motor starts above threshold	Passed
Dashboard	Charts and indicators update	Passed

Tested Temperature Range

22°C – 50°C

Motor Trigger Threshold

28°C

Publish Interval

15 Seconds

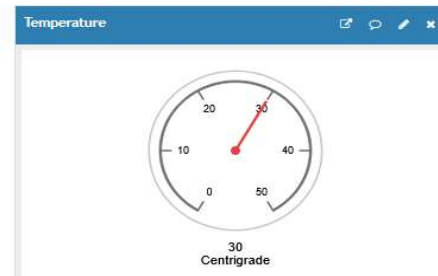
Observed Motor Response

Approximately 2 minutes, due to the ThingSpeak cloud processing and rate-limit behaviour.

RESULT

The system successfully achieved:
Temperature Sensing → MQTT Publishing →
ThingSpeak Cloud → React Rule → ThingHTTP →
MQTT Subscription → Motor Control

This project demonstrates practical experience
in embedded systems, IoT communication,
MQTT, cloud integration, simulation, debugging
and automation.



FUTURE SCOPE OF WORK

The project can be extended with:

- Node-RED integration
- Bidirectional motor control
- Manual ON/OFF control from dashboard
- Additional humidity and pressure sensors
- SMS / email alerts
- AWS IoT Core migration
- Physical Arduino + ESP8266 deployment
- Mobile application using MQTT
- MATLAB-based ThingSpeak data analytics
- Predictive maintenance
- OTA firmware updates
- MQTT SSL/TLS security

THANK YOU!